

Task 4: Clean the Main Architecture (Introducing Multiple Functions)

Introduce **4 helper functions** outside of `main`. We then update `main` to pass its data straight into these functions, keeping the execution flow clean and readable.

```
// Save this entire file as: UnifiedTracker.java
import java.util.Scanner;

public class UnifiedTracker {

    // — TASK 4: FUNCTION 1 (Extracts Text Cleaning Logic from Task 1) —
    public static String formatName(String rawName) {
        String cleanName = rawName.trim();
        if (cleanName.isEmpty()) return "";
        return cleanName.substring(0, 1).toUpperCase() + cleanName.substring(1);
    }

    // — TASK 4: FUNCTION 2 (Extracts Parsing & Casting Logic from Task 2) —
    public static double parseAndCalculateInitial(String textGrade1, String textGrade2) {
        int score1 = Integer.parseInt(textGrade1);
        int score2 = Integer.parseInt(textGrade2);
        return (double) (score1 + score2) / 2;
    }

    // — TASK 4: FUNCTION 3 (Extracts Array Loop Math Logic from Task 3) —
    public static double calculateAverage(int[] gradesArray) {
        int sum = 0;
        for (int score : gradesArray) {
            sum += score;
        }
        return (double) sum / gradesArray.length;
    }

    // — NEW TASK 4: FUNCTION 4 (Replaces and Handles the Task 3 For-Loop Input) —
    public static void fillGradesArray(Scanner inputScanner, int[] scheduleGrades) {
        for (int i = 0; i < scheduleGrades.length; i++) {
            System.out.print("Enter text grade for class #" + (i + 1) + ": ");
            String textGrade = inputScanner.next();
            // Performs the Task 2 Wrapper parsing directly into the array element
            scheduleGrades[i] = Integer.parseInt(textGrade);
        }
    }
}
```

```

// — CLEAN MAIN EXECUTIVE RUNNER —
public static void main(String[] args) {
    Scanner inputScanner = new Scanner(System.in);

    // --- TASK 1 CUMULATIVE REUSE ---
    System.out.println("=== Task 1: String Input & Cleaning ===");
    System.out.print("Enter student name (e.g., ' john doe '): ");
    String rawName = inputScanner.nextLine();

    String formattedName = formatName(rawName); // Function Call 1
    System.out.println("Saved Name: '" + formattedName + "'");

    // --- TASK 2 CUMULATIVE REUSE ---
    System.out.println("\n=== Task 2: Parsing & Casting ===");
    System.out.print("Enter text grade for Math: ");
    String mathText = inputScanner.next();
    System.out.print("Enter text grade for Science: ");
    String scienceText = inputScanner.next();

    double initialAverage = parseAndCalculateInitial(mathText, scienceText); // Function Call 2
    System.out.println("Initial 2-Class Average: " + initialAverage + "%");

    // --- TASK 3 CUMULATIVE REUSE (With Loop Replaced) ---
    System.out.println("\n=== Task 3 & 4: Array Size & Loop Function ===");
    System.out.print("How many total classes do you want to track? ");
    int totalClasses = inputScanner.nextInt();

    // Declare the blank fixed-size array container
    int[] scheduleGrades = new int[totalClasses];

    // Function Call 4: Loops and populates the scheduleGrades array from inside the function
    fillGradesArray(inputScanner, scheduleGrades);

    // --- TASK 4 OUTPUT VERIFICATION ---
    System.out.println("\n=== Task 4: Final Calculated Return Values ===");
    double computedAverage = calculateAverage(scheduleGrades); // Function Call 3
    System.out.println("Clean Function Calculated Average: " + computedAverage + "%");

    inputScanner.close();
}
}

```